

**PROJECT PHASE 2**  
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## ***1.Tables***

```
CREATE TABLE `Teams` (  
    `Team_ID` INTEGER PRIMARY KEY,  
    `Team_Name` VARCHAR(250)  
);
```

```
CREATE TABLE `Players` (  
    `Player_ID` INTEGER PRIMARY KEY,  
    `Player_Name` VARCHAR(50),  
    `Transfer_value` FLOAT  
);
```

```
CREATE TABLE `Plays_at` (  
    `Player_ID` INTEGER,  
    `Team_ID` INTEGER,  
    FOREIGN KEY (`Team_ID`) REFERENCES `Teams`(`Team_ID`),  
    FOREIGN KEY (`Player_ID`) REFERENCES `Players`(`Player_ID`)  
);
```

## 2. Functional Dependencies and BCNF Check

Teams: Team\_ID -> Team\_Name (already BCNF)

Players: Player\_ID -> Player\_Name, Transfer\_value (already BCNF)

Plays\_at: Player\_ID -> Team\_ID (already BCNF)

## 3. Display of Rows

Teams

1 • `select * from teams;`

100% 21:1

Result Grid

| Team_ID | Team_Name       |
|---------|-----------------|
| 1       | Fenerbahçe      |
| 2       | Galatasaray     |
| 3       | Beşiktaş        |
| 4       | Trabzonspor     |
| 5       | Antalyaspor     |
| 6       | Kayserispor     |
| 7       | Adana Demirspor |
| 8       | Kasımpaşa       |
| 9       | Samsunspor      |
| 10      | Konyaspor       |
| NULL    | NULL            |

teams 1

Apply Revert

Action Output

|      | Time     | Action                             | Response           | Duration / Fetch Time  |
|------|----------|------------------------------------|--------------------|------------------------|
| ✓ 23 | 21:15:48 | select * from teams LIMIT 0, 50000 | 10 row(s) returned | 0.00063 sec / 0.000... |

# Players

Trendyol Superleague Explorer

Administration Schemas Query 1 SQL File 3\* SQL File 4\* SQL File 6\* SQL File 7\* SQL File 5\* SQL File 8\* SQL File 9\* SQL File 10\*

SCHEMAS

Filter objects

> sys

> TFF

> Tables

Views

Stored Procedures

Functions

1 • `select * from players;`

Limit to 50000 rows

100% 23:1

Result Grid Filter Rows: Search Edit: Export/Import:

| Player_ID | Player_Name        | Transfer_value |
|-----------|--------------------|----------------|
| ▶ 100     | Edin Dzeko         | 5.5            |
| 101       | Mauro icardi       | 8              |
| 102       | Al Musrati         | 3.5            |
| 103       | Enis Bardhi        | 4.5            |
| 104       | Adam Buksa         | 1              |
| 105       | Majid Hosseini     | 2              |
| 106       | Mario Balotelli    | 1.8            |
| 107       | Haris Hajradinovic | 2.2            |
| 108       | Carlo Holse        | 2.75           |
| 109       | Sokol Cikalleshi   | 1              |
| NULL      | NULL               | NULL           |

players 2 Apply Revert

Action Output

|      | Time     | Action                               | Response           | Duration / Fetch Time   |
|------|----------|--------------------------------------|--------------------|-------------------------|
| ✓ 24 | 21:17:02 | select * from players LIMIT 0, 50000 | 10 row(s) returned | 0.00050 sec / 0.0000... |

Query Completed

Object Info Session

Schema: TFF

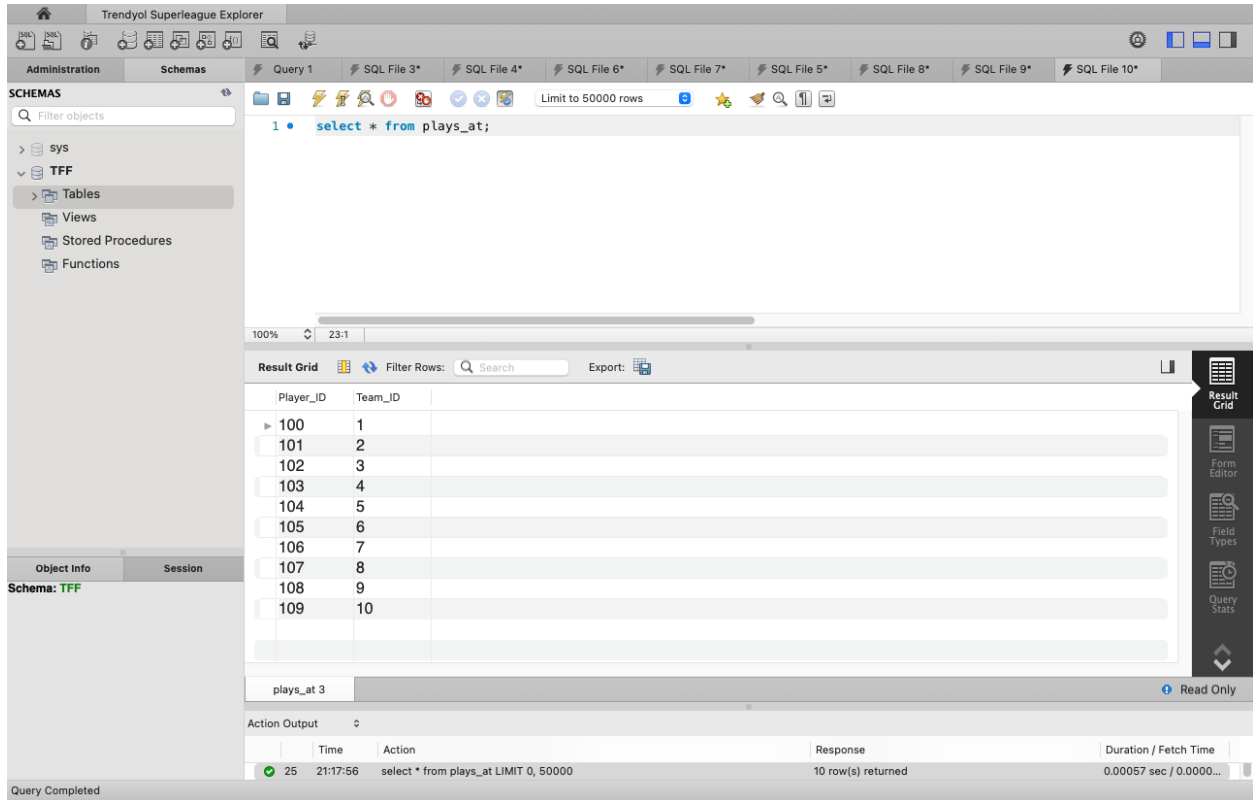
Result Grid

Form Editor

Field Types

Query Stats

# Plays\_at



The screenshot shows the Trendyol Superleague Explorer interface. The left sidebar displays the database schema structure: sys, TFF, Tables, Views, Stored Procedures, and Functions. The main area shows a SQL query: `select * from plays_at;`. Below the query, the 'Result Grid' displays the following data:

| Player_ID | Team_ID |
|-----------|---------|
| 100       | 1       |
| 101       | 2       |
| 102       | 3       |
| 103       | 4       |
| 104       | 5       |
| 105       | 6       |
| 106       | 7       |
| 107       | 8       |
| 108       | 9       |
| 109       | 10      |

Below the result grid, the 'Action Output' section shows the query execution details:

|    | Time     | Action                                | Response           | Duration / Fetch Time   |
|----|----------|---------------------------------------|--------------------|-------------------------|
| 25 | 21:17:56 | select * from plays_at LIMIT 0, 50000 | 10 row(s) returned | 0.00057 sec / 0.0000... |

The status bar at the bottom indicates 'Query Completed'.

## 4.Join

*Query in English:*

*Retrieve the list of players along with the teams they play for.*

*Relational Algebra:*

$\pi(\text{Player\_Name}, \text{Team\_Name})(\text{Plays\_at} \bowtie \text{Players} \bowtie \text{Teams})$



## 5.Group By

*Query in English: Find the number of players in each team.*

The screenshot shows the MySQL Workbench interface. The top toolbar includes icons for file operations and a 'Limit to 50000 rows' dropdown. The left sidebar shows the 'SCHEMAS' tree with 'sys' and 'TFF' (selected) schemas. The main editor displays a SQL query:

```
1 SELECT Teams.Team_Name, COUNT(*) AS PlayerCount
2 FROM Plays_at
3 JOIN Teams ON Plays_at.Team_ID = Teams.Team_ID
4 GROUP BY Teams.Team_Name;
```

Below the query editor is the 'Result Grid' showing 10 rows of data. The columns are 'Team\_Name' and 'PlayerCount'.

| Team_Name       | PlayerCount |
|-----------------|-------------|
| Fenerbahçe      | 1           |
| Galatasaray     | 1           |
| Beşiktaş        | 1           |
| Trabzonspor     | 1           |
| Antalyaspor     | 1           |
| Kayserispor     | 1           |
| Adana Demirspor | 1           |
| Kasımpaşa       | 1           |
| Samsunspor      | 1           |
| Konyaspor       | 1           |

The bottom status bar indicates 'Query Completed' and '10 row(s) returned'.

## 6.Check Constraint

*ALTER TABLE Players*

*ADD CONSTRAINT chk\_transfer\_value\_non\_negative CHECK (Transfer\_value >= 0);*

mysql workbench

Trendyol Superleague Explorer

Administration

Schemas

SCHEMAS

Filter objects

> sys

> TFF

> Tables

Views

Stored Procedures

Functions

Object Info

Session

Schema: TFF

Query 1

SQL File 3\*

SQL File 4\*

SQL File 6\*

SQL File 7\*

SQL File 5\*

SQL File 8\*

SQL File 9\*

SQL File 10\*

Limit to 50000 rows

1

INSERT INTO Players (Player\_ID, Player\_Name, Transfer\_value) VALUES (3, 'Leonardo Bonucci', -50.0);

100%

90:1

Action Output

|    | Time     | Action                                                                           | Response                                                | Duration / Fetch Time |
|----|----------|----------------------------------------------------------------------------------|---------------------------------------------------------|-----------------------|
| 31 | 21:33:26 | INSERT INTO Players (Player_ID, Player_Name, Transfer_value) VALUES (3, 'Leon... | Error Code: 3819. Check constraint 'chk_transfer_val... | 0.0028 sec            |

Query interrupted